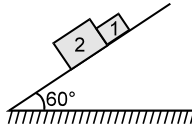


PHYSICS

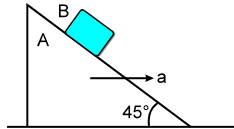
TARGET : JEE- Advanced

CAPS-4
NLM & FRICTION**SCQ (Single Correct Type) :**

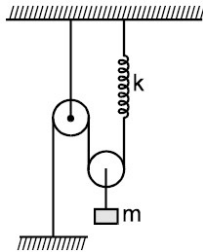
1. In the figure shown if friction co-efficient of block 1 and 2 with inclined plane is $\mu_1 = 0.5$ and $\mu_2 = 0.4$ respectively, then find out the correct statement.



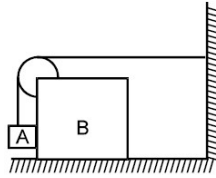
- (A) both block will move together
 (B) both block will move separately
 (C) there is a non-zero contact force between two blocks
 (D) none of these
2. If the coefficient of friction between A and B is μ , the maximum horizontal acceleration of the wedge A for which B will remain at rest w.r.t the wedge is :



- (A) μg (B) $g \left(\frac{1+\mu}{1-\mu} \right)$ (C) $\frac{g}{\mu}$ (D) $g \left(\frac{1-\mu}{1+\mu} \right)$
3. Mass m shown in figure is in equilibrium. If it is displaced further by x and released find its acceleration just after it is released. Take pulleys to be light & smooth and strings light.

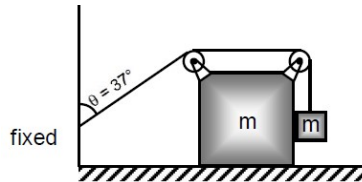


- (A) $\frac{4kx}{5m}$ (B) $\frac{2kx}{5m}$ (C) $\frac{4kx}{m}$ (D) none of these
4. The system is released from rest from the position shown in the diagram. If A and B have mass m and M respectively and all surfaces are smooth, then the normal reaction between A and B is (string and pulley are massless)



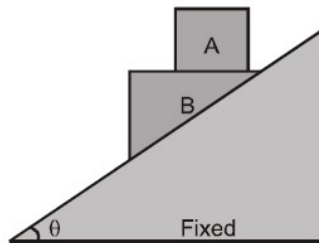
- (A) $\frac{2Mm}{(M+m)}$ (B) $\frac{2Mm}{2(M+m)}$ (C) $\frac{Mmg}{(M+m)}$ (D) zero

5. Consider the arrangement shown in figure. Pulleys and string are ideal. Both blocks have same mass m . The minimum coefficient of friction between bigger block and ground for which both blocks remains in equilibrium is :



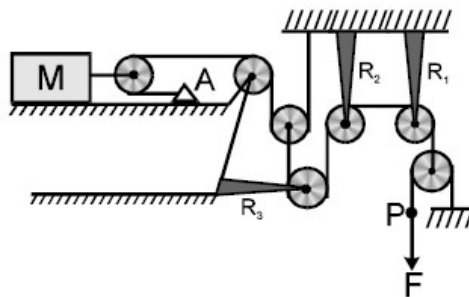
- (A) $\frac{2}{7}$ (B) $\frac{3}{14}$ (C) $\frac{1}{2}$ (D) zero

6. The coefficient of friction between the block A of mass m and wedge B of mass $2m$ is $\mu = \frac{1}{\sqrt{3}}$. The inclined plane is smooth. If the system of blocks A and B is released from rest and there is no slipping between A and B, then $\theta \leq \frac{\pi}{\alpha}$. Find the value of α .



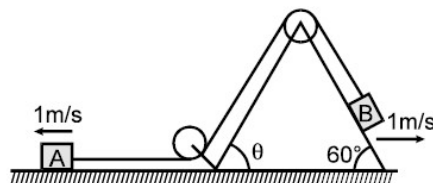
- (A) $\alpha = 5$ (B) $\alpha = 3$ (C) $\alpha = 4$ (D) $\alpha = 6$

7. In the given arrangement, mass of the block is M and the surface on which the block is placed is smooth. Assuming all pulleys to be massless and frictionless, strings to be inelastic and light, R_1 , R_2 and R_3 to be light supporting rods, then acceleration of point P. will be (A is fixed) :

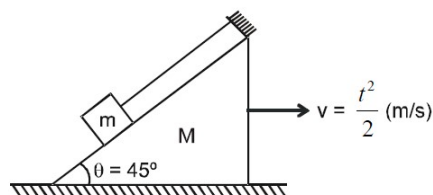


- (A) 0 (B) ∞ (C) $\frac{4F}{m}$ (D) $\frac{2F}{m}$

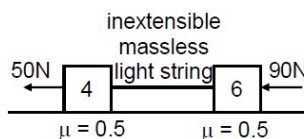
8. A system is shown in the figure. Block A is moving with 1 m/s towards left. Wedge is moving with 1 m/s towards right. Then speed of the block B will be:



- (A) 1 m/s (B) $\sqrt{2}$ m/s (C) $\sqrt{3}$ m/s (D) none of these
9. In the given figure a block of mass m is tied on a wedge by an ideal string as shown in figure. String is parallel to the inclined surface of wedge. All the surfaces involved are smooth. Wedge is being moved towards right with a time varying velocity $v = \frac{t^2}{2}$ (m/s) where t is in sec. Choose the correct option. (use $g = 10 \text{ m/s}^2$)

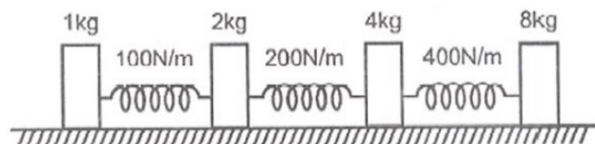


- (A) Block just breaks the contact with wedge after time $t = 20$ sec
 (B) Acceleration of wedge at the instant block just breaks the contact with wedge is 5 m/s^2
 (C) Acceleration of block at the instant block just breaks the contact with wedge is 20 m/s^2
 (D) Speed of block at the instant block just breaks the contact with wedge is 50 m/s
10. Two blocks of masses 4 kg and 6 kg are kept on a rough horizontal surface as shown. Then the acceleration of 6 kg block is.



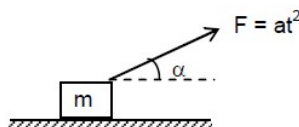
- (A) 7.5 m/s^2 (B) 9 m/s^2 (C) 10 m/s^2 (D) None of these
- MCQ (One or more than one correct) :**

11. Four blocks of masses 1 kg, 2 kg, 4 kg and 8 kg are connected with three ideal springs. The whole system is released from rest on a frictionless horizontal surface. After some time the accelerations of 1 kg and 4 kg are found to be zero while the middle spring is found to be compressed by 0.1 m. At the same instant,

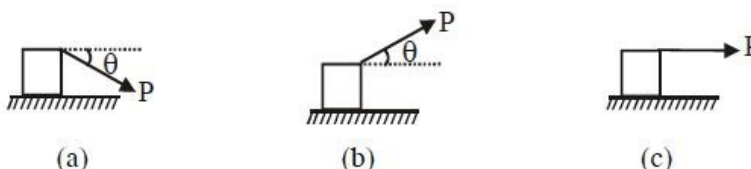


- (A) the acceleration of 8 kg is 2.5 m/s^2 (B) the acceleration of 8 kg is 1 m/s^2
 (C) the acceleration of 2 kg is 10 m/s^2 (D) the acceleration of 2 kg is 5 m/s^2

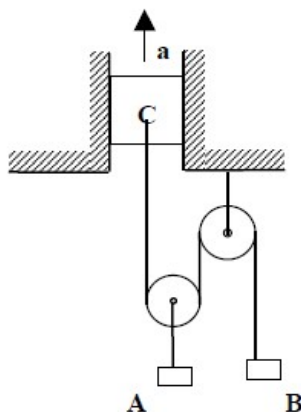
12. At $t = 0$, a force $F = at^2$ is applied to a small body of mass m at an angle α resting on a smooth horizontal plane



- (A) velocity of the body at the moment it breaks off the plane is $\sqrt{\frac{mg^3}{9a \tan^2 \alpha \sin \alpha}}$
- (B) the distance travelled by the body before breaking off the plane is $\frac{mg^2}{12a \sin \alpha \tan \alpha}$
- (C) Its acceleration at the time of breaking off the plane is $g \cot \alpha$.
- (D) Time at which it breaks off the plane is $mg \sqrt{\frac{mg}{a \sin \alpha}}$
13. Figure shows three blocks on a rough surface under influence of a force P of same magnitude in all the three cases. Coefficient of friction is same between each block and ground. What possible relation holds between magnitudes of normal reaction and friction forces. [Assume that blocks do not overturn about edge, f_a, f_b & f_c are frictional forces & N_a, N_b & N_c are reactions]



- (A) $N_a > N_c > N_b$ (B) $f_a > f_c > f_b$ (C) $f_c > f_a = f_b$ (D) $N_c > N_a = N_b$
14. The block C shown in the figure is ascending with an acceleration $a = 3\text{ m/s}^2$ by means of some motor not shown here. bodies A and B have masses 10 kg and 5 kg respectively (assume pulleys are massless and friction is absent everywhere).

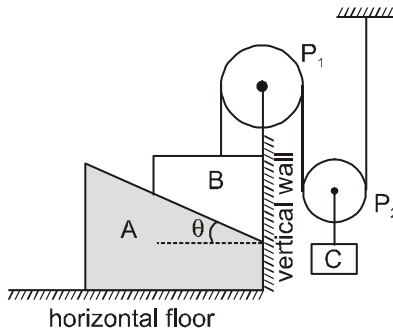


- (A) The acceleration of block A is 1 m/s^2 upwards
- (B) The acceleration of block A is 1 m/s^2 downwards
- (C) The acceleration of block B is 1 m/s^2 upwards
- (D) The acceleration of block B is 1 m/s^2 downwards

Comprehension Type Question:

In the figure shown P_1 and P_2 are massless pulleys. P_1 is fixed and P_2 can move. Masses of A, B and C are $\frac{9m}{64}$, $2m$ and m respectively. All contacts are smooth and the string is massless.

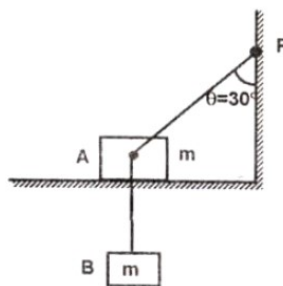
$$\theta = \tan^{-1} \left(\frac{3}{4} \right). \text{ (Take } g = 10 \text{ m/s}^2 \text{)}$$



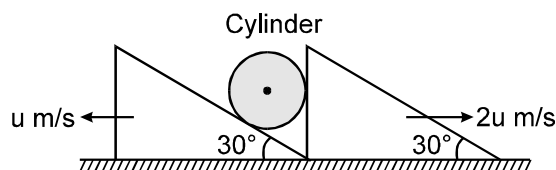
15. The ratio of magnitude of accelerations of blocks A and C
- (A) $\frac{3}{2}$ (B) $\frac{4}{3}$ (C) 2 (D) $\frac{8}{3}$
16. The acceleration of block C is
- (A) 1 m/s^2 (B) 3 m/s^2 (C) 4 m/s^2 (D) 8 m/s^2
17. The tension in string connecting pulley P_2 and block C is (Take $m = 1 \text{ kg}$)
- (A) 3 N (B) 4.5 N (C) 6.5 N (D) 13 N

Numerical based Questions :

18. Block A is on a frictionless horizontal table. A massless inextensible string fixed at one end passes over a smooth nail fixed with the block A. The other end of the string is connected to block B of mass m . Initially the block B is held at rest so that $\theta = 30^\circ$. What will be the magnitude of acceleration of block B just after it is released (in m/s^2 take $g = 10 \text{ m/s}^2$).



19. System is shown in the figure. Assume that cylinder remains in contact with the two wedges. Find the velocity of cylinder.



20. In the situation shown in figure, for what value of minimum horizontal force F (in Newton), sliding between middle and lower block will start? (Take $g = 10 \text{ m/s}^2$)

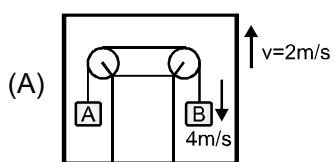


Matrix Match Type :

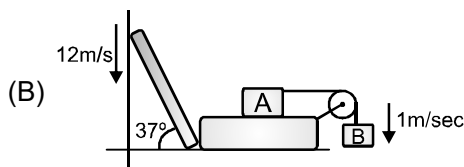
21. In column-I four situations are given in which two blocks A and B are presents at a given instant and $\vec{v}_A$ is velocity of block A, $\vec{v}_B$ is velocity of block B then match column-I with column-II. (all velocities are given with respect to ground).

Column-I

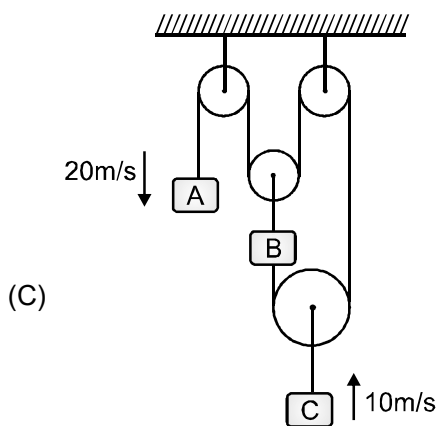
Column-II



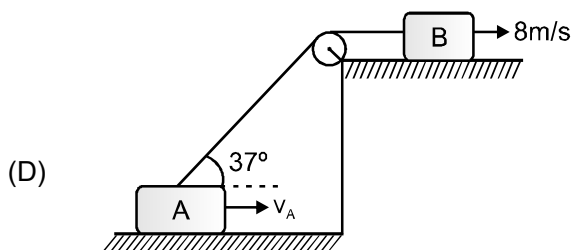
(p) speed of A is greater than B



(q) speed of B is greater than A



(r) $\vec{v}_A \cdot \vec{v}_B = \text{positive}$

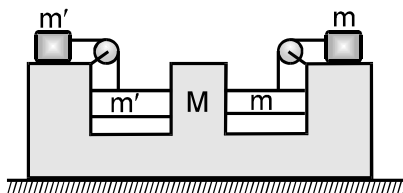


(s) $\vec{v}_A \cdot \vec{v}_B = \text{negative}$

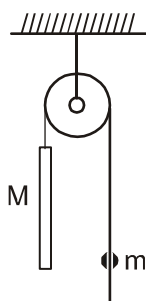
(t) $\vec{v}_A \cdot \vec{v}_B = 0$

Subjective Type Questions :

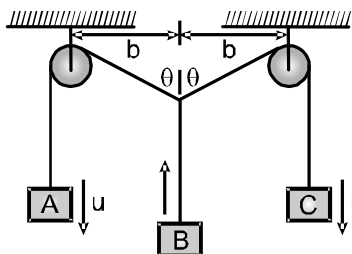
22. Neglecting friction every where, find the acceleration of M . Assume $m > m'$.



23. A small mass slides down an inclined plane of inclination θ with the horizontal. The co-efficient of friction is $\mu = \mu_0 x$ where x is the distance through which the mass slides down and μ_0 , a constant. Then find
- Maximum speed of particle
 - How much distance it will cover to get that maximum speed
24. In the arrangement shown in Fig. the mass of the rod M exceeds the mass m of the ball. The ball has an opening permitting it to slide along the thread with some friction. The mass of the pulley, mass of the string and the friction in its axle are negligible. At the initial moment the ball was located opposite the lower end of the rod. When set free, both bodies began moving with constant accelerations. Find the friction force between the ball and the thread if t seconds after the beginning of motion the ball got opposite the upper end of the rod. The rod length equals ℓ .



25. In the figure shown the blocks A & C are pulled down with constant velocities u . Find the acceleration of block B.



ANSWER KEY_CAPS_04

- | | | | | |
|---|-------------------------------------|-------------------------------------|---|----------|
| 1. (B) | 2. (B) | 3. (C) | 4. (D) | 5. (B) |
| 6. (D) | 7. (C) | 8. (C) | 9. (D) | 10. (C) |
| 11. (AC) | 12. (ABCD) | 13. (ABC) | 14. (AC) | 15. (D) |
| 16. (B) | 17. (D) | 18. (0002) | 19. $\sqrt{7}u$ m/s | 20. 30 N |
| 21. (A) – p,s ; (B) – p,t ; (C) – p,t ; (D) – p,r | | | 22. $a = \frac{(m - m')g}{2M + 3m + 3m'}$ | |
| 23. (a) $v_{\max} = \sqrt{\frac{g \sin \theta \tan \theta}{\mu_0}}$ | (b) $x = \frac{\tan \theta}{\mu_0}$ | 24. $F_{fr} = 2mM\ell / (M - m)t^2$ | 25. $\frac{u^2}{b} \tan^3 \theta$ | |